

Geospatial Heterogeneity in Inflation: A Market Concentration Story*

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Abstract

We study spatial heterogeneity in inflation across regions with different income levels and the role of retailer market structure. Using NielsenIQ Retail Scanner data, we document that, from 2006 to 2020, poorer MSAs experienced annual food inflation 0.46 percentage points higher than in richer MSAs, implying an 8.8 percentage point cumulative gap. Poorer areas have fewer products, fewer retailers, and higher market concentration. To identify causal effects, we exploit the 2014-2015 avian influenza outbreak and show that higher local retail concentration amplifies the pass-through of cost shocks. We develop a heterogeneous-firm model with customer capital accumulation that rationalizes these patterns.

JEL Code: E31, L11, R12

Keywords: inflation; retailer market structure; market concentration; market power; pass-through of cost shocks; spatial inequality

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1 Introduction

Inflation is a key economic indicator with broad implications for economic growth, stability, and household well-being. However, it is typically measured and analyzed at the national level.¹ As a result, both the literature and policy discussions often overlook heterogeneity in inflation rates across regions, potentially masking important local disparities.

Understanding local variation in food inflation is crucial for several reasons. Households across regions face different price changes and adjust their consumption accordingly. Food markets are also more localized and segmented than many other sectors, with local market structures playing an important role in price variation.² Moreover, food is a necessity and accounts for a disproportionately large share of spending for low-income and vulnerable households.³ Taken together, these factors suggest that spatial variation in food inflation can have meaningful implications for consumer welfare and spatial inequality, yet this dimension remains underexplored in the literature.

In this paper, we address this gap by documenting spatial heterogeneity in food inflation and examining the role of retailer market structure in driving this variation and its aggregate implications. Using granular NielsenIQ Retail Scanner data at the 12-digit universal product code (UPC) level, we construct price indexes for disaggregated personal consumption expenditure (PCE) food items at the metropolitan statistical area (MSA) level and document new facts on spatial heterogeneity in food inflation and retailer market structure. We then develop a novel identification strategy to establish the causal relationship between retailer market concentration and inflation, and build a model of heterogeneous retailers with customer capital to account for the mechanism.

We document several new stylized facts. First, we find that food inflation rates vary systematically across regions by income level: poorer MSAs experienced higher inflation than wealthier MSAs between 2006 and 2020, with a cumulative gap of approximately 8.8 percentage points between the bottom and top income deciles. This pattern holds across both disaggregated and

¹The Bureau of Labor Statistics (BLS) provides regional price indexes, but only for a limited subset of large metropolitan areas. They are restricted to 23 Core Based Statistical Areas (CBSAs), which are mostly rich areas, including New York–Newark–Jersey City, Los Angeles–Long Beach–Anaheim, Chicago–Naperville–Elgin, San Francisco, Boston, Washington. See more details at <https://www.bls.gov/cpi/regional-resources.htm>.

²In the NielsenIQ Consumer Panel, we find that 92% of households purchase food exclusively within their home MSAs. See Online Appendix A for details.

³Schanzenbach et al. (2016) report that low-income households spend nearly 20% of their total expenditure on food, compared to 13% for middle-income households and an even smaller share for high-income households, based on the Consumer Expenditure Survey.

aggregated food categories and remains robust to two key restrictions: i) imposing a “common goods rule” that limits the sample goods to UPCs sold in all ten income deciles, and ii) applying uniform expenditure weights across deciles when constructing the index. Together, these results indicate that the documented inflation gap does not merely reflect differences in consumption baskets or product expenditure weights across regions.

Second, we find substantial regional variation in product varieties and retailer market structure. Richer areas offer more varieties of goods (UPCs) and have more stores and retail chains.⁴ We also find systematic differences in retailer market structure across regions by income level. In NielsenIQ Retail Scanner data, we classify retailers as large or small based on total sales or national store counts—defining large (small) retailers as those in the top (bottom) decile. Poorer MSAs have a higher share of large retailers and a lower share of small ones, while the opposite holds in richer MSAs.⁵ We also find that retail sales are lower and more concentrated in poorer areas.

Finally, descriptive OLS regressions reveal a positive association between retailer market concentration—measured by the sales-based Herfindahl–Hirschman Index (HHI)—and inflation at the MSA level. To move beyond a descriptive association between inflation and market concentration, we exploit the 2014-2015 bird flu (avian influenza) episode in the egg market, which provides plausibly exogenous variation in exposure across regions. Using a triple-difference framework, we compare inflation across MSAs with varying levels of market concentration. We find that MSAs with higher concentration experienced significantly higher inflation following the shock. Using a back-of-the-envelope calculation, we find that the egg inflation gap between the bottom and top income deciles from 2014Q4 to 2015Q3 was 14.5 percentage points, with approximately 64% of this gap explained by differences in market concentration.

Furthermore, we find persistent pass-through of shocks over time. In MSAs with higher market concentration, inflation rose more during the inflationary phase of the shock (from 2014Q4 to 2015Q3), while prices adjusted downward less during the subsequent deflationary phase (after 2015Q3). The absence of offsetting deflation during the deflationary period suggests that temporary shocks can have persistent effects on regional inflation, contributing to long-run spatial disparities in

⁴Typically, the set of UPCs available in poorer regions is a subset of those found in richer areas. As a result, imposing the common goods rule across income deciles disproportionately reduces the set of UPCs in richer areas.

⁵These patterns are robust when using the Business Dynamics Statistics (BDS), where retailer size is defined by employment. We define large retailers as those with 500 or more employees and small retailers as those with 19 or fewer employees.

prices and potentially amplifying real income inequality. These patterns are not driven by differences in consumption baskets, retailer composition, or local cost factors. Analogous pass-through patterns in the coffee market provide further support for the role of market concentration.

To rationalize the mechanism underlying our empirical findings, we develop a model of heterogeneous firms with customer capital that links market concentration to pass-through. We build on [Atkeson and Burstein \(2008\)](#) and incorporate customer capital accumulation following [Foster et al. \(2016a\)](#). In the model, firms with larger sales shares obtain a higher level of customer capital and charge higher markups. Consequently, sufficiently large firms pass through cost shocks more aggressively, as the marginal loss of customer capital is smaller and demand becomes less elastic with respect to own-price changes. Markets with higher concentration have higher aggregate markups, exhibit larger increases in aggregate prices following a cost shock, and sustain elevated price levels afterward. This mechanism implies that market concentration amplifies welfare losses from market power in the presence of cost shocks, so policies that shape local retailer market structure can affect both cost pass-through and consumer welfare.

These findings have important distributional and spatial policy implications because retail food markets are highly localized. In poorer regions, where these markets are more segmented, higher food inflation has a disproportionate effect on consumer welfare. This effect is particularly pronounced for vulnerable and less mobile households in low-income communities, which devote a larger share of expenditure to food and face limited access to alternative retailers and substitute goods.⁶ As a result, the interaction between local market segmentation and concentrated retail market structures disproportionately burdens consumers in poorer regions.

Moreover, when real income is evaluated using locally constructed food price indexes, real income inequality widens relative to measures based on national price indexes.⁷ National price indexes systematically understate inflation in poorer regions. This highlights a key limitation of national inflation measures and underscores the importance of accounting for regional variation in both inflation and market structure when evaluating welfare and designing policies.

Related Literature. This paper contributes to several strands of literature. First, our work relates to

⁶In part, this reflects limited geographic mobility, as migration rates have declined since the 1980s ([Kerns-D'Amore et al., 2022](#)).

⁷This is consistent with [Beck and Jaravel \(2020\)](#) and [Jaravel \(2024\)](#), who document biases in standard aggregate price indexes arising from non-homothetic preferences and heterogeneous inflation across households.

the literature on inflation heterogeneity across different groups. [Hobijn and Lagakos \(2005\)](#) and [Hobijn et al. \(2009\)](#) document inflation differences across demographic groups in the Consumer Expenditure Survey (CEX) data. [Kaplan and Menzio \(2015\)](#) and [Kaplan and Schulhofer-Wohl \(2017\)](#) show that low-income and old households face higher inflation even for the same bundle of goods in NielsenIQ data.⁸ [Jaravel \(2018\)](#) emphasizes the role of product innovation and segmented consumption goods. [Argente and Lee \(2021\)](#) find that high-income households face lower inflation during recessions by substituting toward lower-quality goods, while [Becker \(2024\)](#) shows that such demand shifts raise markups on cheaper goods purchased by poor households. [Handbury \(2021\)](#) and [Molloy \(2024\)](#) further highlight the roles of income-specific tastes and housing, respectively. This literature focuses primarily on household-level heterogeneity and consumer-side mechanisms, such as basket composition, preferences, and search behavior. In contrast, we document regional heterogeneity in inflation across MSAs with different income levels and identify retailer market structure as a novel source of inflation variation. This spatial variation can provide a tractable avenue for policy intervention because differences across locations are observable and can be addressed through existing policy tools, such as place-based transfers, competition policy, and local market regulation.

Another closely related strand of literature examines retailer market concentration and market power. A large body of work documents rising retailer concentration and the growing role of national chains ([Jarmin et al., 2009](#); [Haltiwanger, 2012](#); [Hortaçsu and Syverson, 2015](#); [Foster et al., 2016b](#); [Cao et al., 2024](#); [Smith and Ocampo, 2025](#)). In contrast, [Rossi-Hansberg et al. \(2021\)](#) and [Benkard et al. \(2021\)](#) find declining concentration in local markets and narrowly defined product markets, respectively. Other papers estimate retailer markups and their heterogeneity, linking them to city size ([Hottman, 2017](#)), local housing prices ([Stroebe and Vavra, 2019](#)), rising product variety ([Brand, 2021](#)), household search behavior ([Sangani, 2022](#)), changes in marginal costs and price sensitivity ([Döpper et al., 2025](#)), and the rise in concentration of household shopping ([Leung and Li, 2026](#)).⁹ [Jaravel \(2018\)](#) and [Jaravel \(2021\)](#) show that firms disproportionately innovate in products with higher demand from rich households, leading to greater product entry and stronger competition in markets serving richer consumers. Our paper contributes to this literature by documenting

⁸[O’Flaherty \(2026\)](#) documents substantial dispersion in relative price changes across products and shows that some within-product dispersion across households also exists.

⁹Similarly, [Franco \(2024\)](#) finds that markups are lower in larger cities using the Census micro-level data.

systematic variation in retailer market concentration across MSAs with different income levels and by establishing a causal link between concentration and inflation.

Our paper is also related to the literature on food deserts and local retail access, which emphasizes spatial disparities in the availability of grocery stores and healthy foods (Bitler and Haider, 2011; Allcott et al., 2019). We highlight a complementary mechanism operating through local market structure: retail environments with fewer competitors tend to be more concentrated, which can increase markups and amplify the pass-through of cost shocks, contributing to systematically higher food inflation.

Our paper also contributes to the literature on the pass-through of shocks to retail prices and inflation. Prior work studies pass-through of demand shocks (Arcidiacono et al., 2020; Gagnon and López-Salido, 2020; Handbury and Moshary, 2021) and supply shocks, including tax changes (Cawley et al., 2018, 2020; Baker et al., 2020; Butters et al., 2022), wholesale cost fluctuations (Nakamura and Zerom, 2010), and commodity price changes in coffee (Sangani, 2024). We differ from this literature by exploiting the 2014-2015 avian influenza outbreak, which generated a sharp, temporary, and exogenous supply shock in the US egg market with regional variation in exposure. This setting allows us to identify how pass-through varies systematically across MSAs with different levels of market concentration. We are among the first to provide causal evidence on inflation heterogeneity by linking regional differences in market concentration to differential pass-through of shocks. We further propose a customer-capital mechanism underlying this pattern: larger firms in concentrated markets hold greater market power and internalize the dynamic value of their customer base differently, which alters their incentives to pass through cost shocks.¹⁰

Lastly, our study contributes to the literature using item-level data to construct more accurate inflation measures (Ehrlich et al., 2023), which can outperform official indexes in capturing true cost-of-living changes (Handbury et al., 2013). Related efforts include Beck and Jaravel (2020), who correct biases in standard cross-country price indexes by constructing a scanner-data-based index that accounts for non-homothetic preferences, product variety, and taste heterogeneity; Jaravel and Lashkari (2024), who derive a correction to real consumption (welfare) that accounts for heterogeneity in consumption baskets and inflation; and Jaravel (2024), who constructs Distributional

¹⁰This mechanism is closely related to Kleshchelski and Vincent (2009), Foster et al. (2016a), and Paciello et al. (2019).

Consumer Price Indexes (D-CPIs) based on heterogeneous inflation rates across demographic groups and shows that real inequality grows faster under D-CPIs than under official CPIs. In a similar vein, we construct regional price indexes at the MSA level to better capture spatial heterogeneity in inflation.¹¹ In contrast, official indexes are typically national and expenditure-weighted, or regional but limited to larger and wealthier areas. As a result, poorer regions tend to be underrepresented due to lower aggregate spending and more limited coverage.¹² Our regional indexes provide a more representative measure of local inflation and, therefore, real income inequality. We find that poorer areas experience higher food inflation, which widens real income disparities relative to nominal differences.¹³

The rest of the paper is organized as follows. Section 2 describes the data and key measures; Section 3 presents stylized facts on spatial heterogeneity in food inflation and retailer market structure; Section 4 outlines the empirical strategy for causal inference and presents the main empirical results; Section 5 provides a theoretical framework and the underlying mechanism; Section 6 discusses potential policy implications; and Section 7 concludes.

2 Data and Measures

The primary dataset we use to analyze heterogeneous inflation rates across regions is the NielsenIQ Retail Scanner dataset. The NielsenIQ dataset enables us to measure inflation rates and retailer market structure across regions using data on food sales, prices, and store locations.¹⁴

2.1 NielsenIQ Retail Scanner

Our analysis uses the Retail Measurement Services (RMS) dataset provided by the Kilts Center at Chicago Booth. This dataset includes weekly pricing, volume, and store merchandising data from

¹¹Our index construction is closely related to that in [Handbury and Weinstein \(2014\)](#) and [Diamond and Moretti \(2021\)](#). Unlike [Handbury and Weinstein \(2014\)](#), we do not correct for either heterogeneity bias or variety bias because our objective is to capture observed differences in households’ consumption baskets.

¹²This concern echoes [Martin \(2024\)](#), who highlights that expenditure-weighted indexes may systematically understate inflation in poorer areas—especially when price dynamics vary across regions.

¹³This contrasts with [Moretti \(2013\)](#) and [Diamond and Moretti \(2021\)](#), who document that real inequality is lower than nominal inequality when taking into account variations in goods coverage, geographic scope, or cost-of-living. Our analysis focuses on food prices using highly granular UPC-level data.

¹⁴As a secondary dataset, we use the Business Dynamics Statistics (BDS) from the U.S. Census Bureau, which covers the universe of firms in the retail sector.

over 100 retail chains across U.S. markets, covering more than 40,000 individual stores. Total sales in the NielsenIQ RMS sample exceed \$200 billion annually, representing 50% of grocery store sales, 55% of drug store sales, 32% of mass merchandiser sales, and 2% of convenience store sales.

A key advantage of this dataset is that it contains detailed information at the finest product level: 12-digit universal product codes (UPCs) that uniquely identify specific goods.¹⁵ The dataset contains over 2.6 million UPCs. Furthermore, NielsenIQ classifies UPC-level goods into 10 departments, 110 product groups, and over 1,000 product modules.

We further use a concordance provided by the U.S. Bureau of Labor Statistics (BLS) that maps NielsenIQ product modules to BLS entry-level items (ELIs).¹⁶ These ELIs then map to Personal Consumption Expenditure (PCE) disaggregated categories. Our analysis focuses on the food sector, which is identified as the aggregation of 21 PCE food categories, spanning from 2006Q1 to 2020Q3. The following lists these 21 categories: Bakery, Beef and Veal, Beer, Cereal, Coffee, Dairy, Eggs, Fats and Oil, Fish, Fruits, Milk, Other Foods, Other Meats, Pork, Poultry, Processed Fruits and Vegetables, Soda, Spirits, Sugar and Sweets, Vegetables, and Wine.

To construct our main dataset from NielsenIQ, we start with the raw data at the weekly-store-UPC level and link it to personal income data at the MSA level from the U.S. Bureau of Economic Analysis (BEA) based on store location information in NielsenIQ.¹⁷ We then define income deciles by the cross-time average of MSA-level income per capita. Table 1 reports examples of MSAs in the top, median, and bottom income deciles.

The weekly price data are first aggregated into months according to the National Retail Federation (NRF) calendar and then into quarters by grouping three consecutive NRF months.¹⁸ Using the concordance between product modules and PCE food categories, we identify the food sector in NielsenIQ. Finally, to measure retailer market structure and the degree of competition, we link store identifiers to retail chain identifiers using a crosswalk provided by NielsenIQ.

Our main analysis is at the MSA (or income decile)-food category-quarter level. We generate price indexes, the Herfindahl-Hirschman Index (HHI) of sales concentration, the share of top

¹⁵For example, two cans of Campbell's tomato soup in different sizes would be classified as two different UPCs.

¹⁶ELIs are the most granular complete mutually exclusive classification of CPI items produced by the BLS. We were provided this concordance as part of the Re-Engineering Statistics using Economic Transactions (RESET) project.

¹⁷Our baseline uses retailers' MSA locations in NielsenIQ. To address cross-MSA shopping concerns, we use NielsenIQ Consumer Panel data to examine its prevalence and compare income deciles defined by store MSA versus household MSA; see Online Appendix A.

¹⁸The NRF calendar typically starts in early February and ends around the end of January of the following year.

Table 1: Examples of MSAs by Decile

Decile	MSA
1 (lowest)	El Paso (TX), Albany (GA), Yuma (AZ), etc.
5	Knoxville (TN), Panama City (FL), Binghamton (NY), etc.
10 (highest)	New York (NY), Washington (DC), Boston (MA), etc.

Note: The table provides some examples of MSAs located in the deciles 1, 5, and 10. These deciles are time invariant in our setting and are based on income per capita data from the BEA, averaged over the period 2006-2020.

retailers, and other statistics associated with market structure for each MSA (or income decile)–food category–quarter combination.

2.2 Main Measures

2.2.1 Price Indexes

To measure and compare food inflation rates across regions with different income levels, we construct price indexes either at the MSA level or by income deciles, using UPC-level data from NielsenIQ. As a starting point, we adopt a traditional price index to measure inflation, the log geometric Laspeyres price index, which is calculated as follows:

$$\ln \Psi_{mt}^G = \sum_{k \in \mathbb{C}_{m,t-1,t}} \omega_{mkt} \ln \frac{p_{mkt}}{p_{mkt-1}}, \quad (1)$$

where ω_{mkt} represents the weight assigned to product k in quarter t for MSA (or income decile) m , and we use lagged expenditure shares as weights, e.g., $\omega_{mkt} = s_{mkt-1}$. The set $\mathbb{C}_{m,t-1,t}$ includes all “continuing” goods, defined as products sold in both periods t and $t - 1$ in MSA (or income decile) m .¹⁹

We also construct two alternative price indexes: i) one by restricting the sample to UPCs that are sold in all ten income deciles in both consecutive quarters $t - 1$ and t , referred to as “common goods,” and ii) another by restricting the sample to common goods and applying common weights across deciles. Specifically, we impose $\omega_{mkt} = \omega_{kt}$ for all deciles m , where the weights are fixed to the lagged expenditure shares in the bottom income decile. This approach accounts for the

¹⁹To assess the representativeness of NielsenIQ, we compare our price indexes with the official CPI series for the MSAs available in the CPI data. See more details in Online Appendix B.

fact that consumption baskets can differ systematically between income groups, as documented in [Jaravel \(2018\)](#), and consequently between regions with different income levels. By focusing on a common set of goods and applying common weights, we move closer to isolating regional inflation differences that are not driven by variation in the composition of consumption baskets.

In addition, we conduct a robustness test using alternative price indexes to address potential biases associated with our baseline measure, such as substitution bias or chain drift.²⁰ See more details in Online Appendix [C.1](#).

2.2.2 Retailer Market Structure

In NielsenIQ data, we use store and retailer codes, together with geographic information for each store, to identify stores and retailers and trace their ownership structures across regions and over time. We define the size of retailers based on either total sales or the number of stores they own at the national level. We classify large chains as those in the top decile and small chains as those in the bottom decile of the national size distribution. We use the local HHI of retailer sales as our main measure of market concentration in each MSA.²¹ Alternatively, we use the sales share of the top one or three retailers within an MSA. Table [2](#) provides summary statistics for the main sample.

3 Descriptive Evidence

3.1 Spatial Heterogeneity in Retail Prices and Inflation

Figure [1](#) presents the geometric Laspeyres price index for aggregate food, constructed from the NielsenIQ Retail Scanner data by income decile. We report results for the first (poorest), fifth, and tenth (richest) income deciles, alongside the official PCE food price index for comparison. The base quarter is set to 2006Q2.²²

²⁰Fixed-basket price indexes do not fully account for demand responses arising from consumers substituting across differentiated goods, and chained price indexes may be subject to chain drift.

²¹As emphasized by [Covarrubias et al. \(2020\)](#) and [Syverson \(2019\)](#), market concentration remains an imperfect proxy for market power. This consideration motivates our reliance on natural experiments for causal identification (Section [4](#)) and on a theoretical framework (Section [5](#)) to discipline the interpretation of our results.

²²Price indexes are constructed using price and expenditure information from both periods t and $t - 1$. Therefore, 2006Q2 is the first quarter for which we are able to estimate a price index.

Table 2: Summary Statistics of the Main Sample (MSA-Quarter Level)

	Mean (SD)		Mean (SD)
Income per capita (\$ thousands)	42.49 (9.29)	Share of Large Chains (Top Decile by Sales)	0.357 (0.11)
Sales (\$ millions)	207.23 (366.62)	Share of Small Chains (Bottom Decile by Sales)	0.016 (0.04)
Population Share	0.005 (0.01)	Share of Large Chains (Top Decile by Store #)	0.619 (0.16)
Sales Share of Chains	0.117 (0.04)	Share of Small Chains (Bottom Decile by Store #)	0.008 (0.03)
Number of Chains	9.74 (3.71)	Market Concentration (HHI)	0.416 (0.18)
Number of Stores	193.84 (251.38)	Market Concentration (CR1)	0.534 (0.19)
Number of UPCs	49180.16 (18954.19)	Market Concentration (CR3)	0.817 (0.12)
Observations	11,100	Observations	11,100
Number of MSAs	185	Number of MSAs	185
Number of Quarters	60	Number of Quarters	60

Note: The table provides the summary statistics of the main MSA-level sample for the aggregate food and beverages in NielsenIQ. The population share is defined as the fraction of total population in an MSA. The sales share of chains is the average retailer-level sales share within an MSA, and the number of chains, stores, and UPCs is defined as the average number of chains, stores, and UPCs within an MSA. Large (small) chains are defined as those in the top (bottom) decile of the national distribution of total sales or store counts in a given quarter, and we compute the share of large or small chains within an MSA. Market concentration is measured using either the HHI of chain-level sales or the sales share of the top one or three retailers (CR1 or CR3) within an MSA.

The general trend in Figure 1 indicates that the poorest decile (“Decile 1”) experiences consistently higher food price growth than the richer deciles (“Decile 5” and “Decile 10”). On average, annualized food inflation is approximately 0.46 percentage points higher in the bottom income decile compared to the top income decile. Over the sample period, this corresponds to a cumulative inflation gap of nearly 8.8 percentage points between the poorest and richest deciles.

This pattern continues to hold even when we restrict the sample to the set of common goods sold across all income deciles in consecutive quarters $t - 1$ and t , and when we apply uniform sales weights. These results are shown in the left and right panels in Figure 2, respectively. These results suggest that the observed variation in price growth across income deciles is not primarily driven by

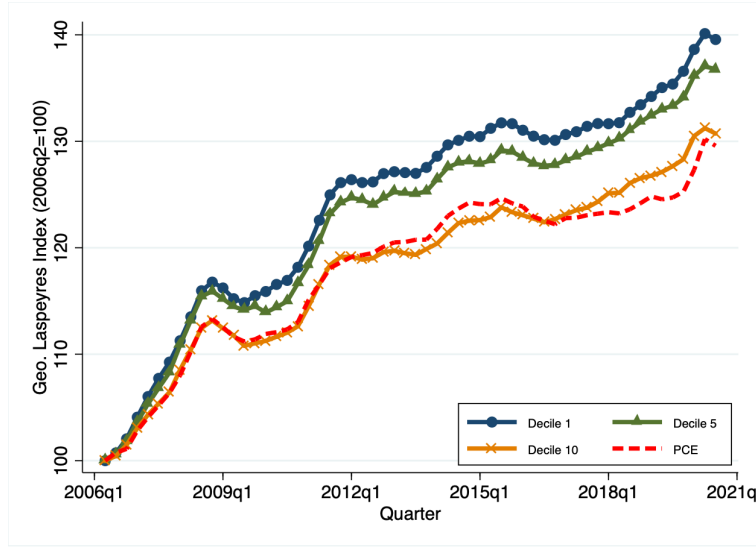


Figure 1: Price Index for Aggregated Food

Notes: This figure represents the chained geometric Laspeyres price index constructed using NielsenIQ Scanner data (solid lines) as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for aggregate food and beverages. The sample period runs from 2006Q2 to 2020Q3, and all series are normalized to the initial quarter. Each solid line corresponds to an income-per-capita decile of MSAs, where decile 1 includes MSAs with the lowest income per capita and decile 10 includes those with the highest income per capita. NielsenIQ UPCs are mapped to the PCE definition of food purchased for off-premises consumption using a product module concordance provided by the U.S. Bureau of Labor Statistics.

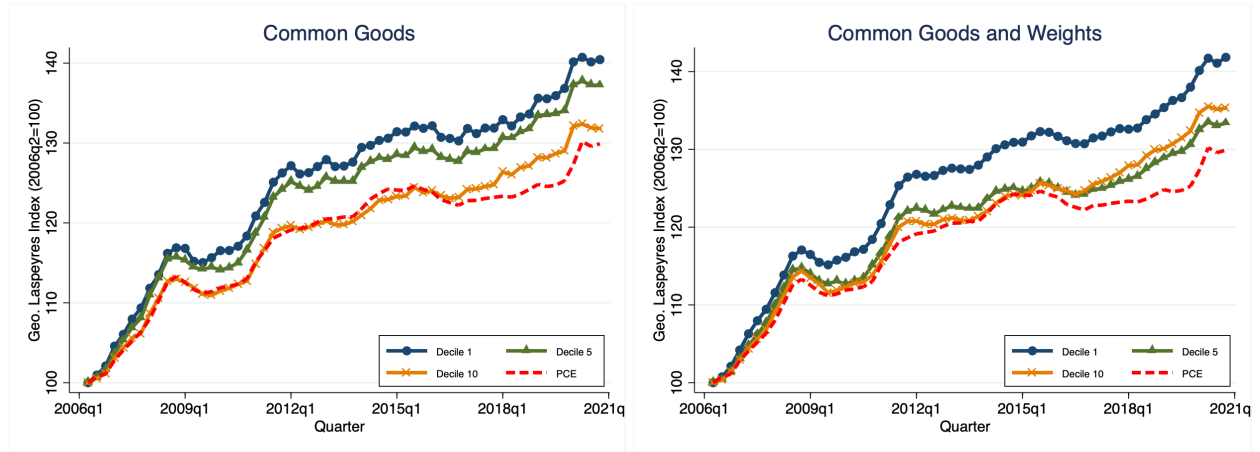


Figure 2: Price Index for Aggregated Food (Common Goods and Weights)

Notes: This figure represents the chained geometric Laspeyres price index constructed using NielsenIQ Retail data (solid lines) as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for aggregate food and beverages. All descriptions remain the same as before except that the solid lines constructed in NielsenIQ are restricted to the set of goods present across all ten deciles in quarters $t - 1$ and t (named “common goods”) in the left panel, and are restricted to these common goods and further based on the same sales weights in decile 1 in the right panel.

differences in consumption baskets, their composition, or consumer preferences across regions.

These patterns are also robust to alternative price indexes, including chained and direct (bilateral) Tornqvist indexes, a weighted time-product dummy (WTPD) index, a demand-system-consistent Sato-Vartia index, and an MSA-level index (Online Appendix C.1). These patterns also hold across disaggregated PCE food categories (Online Appendix C.2).

Lastly, the official PCE series aligns more closely with the NielsenIQ series for the highest income decile than for any other decile. This suggests that aggregate price indexes more closely track inflation in higher-income areas and understate inflation in lower-income areas.²³

These differences in food inflation are consistent with several findings in the literature. First, DellaVigna and Gentzkow (2019) show that retailers set nearly uniform prices across locations for a given product, which does not exclude spatial inflation differences arising from differences in the set of retailers and products available across markets.²⁴ Consistent with this mechanism, Kim and Navarrete (2026) document systematic differences in the composition of products and retailers across MSAs along the income distribution. Finally, our findings relate to Butters et al. (2022), who show that local cost shocks are passed through locally rather than uniformly at the national level, suggesting that local market conditions can contribute to spatial differences in price dynamics.

3.2 Spatial Heterogeneity in Retailer Market Structure

We examine retailer market structure across regions with different income levels. Using the MSA-level NielsenIQ sample, we compute summary statistics by decile of income per capita. Table 3 presents these statistics. Higher-income areas have more retailers and stores, greater total sales, and lower average sales shares per retailer. In contrast, poorer areas offer fewer UPCs and allocate a larger share of total consumption—both in quantities and expenditure—to the set of common goods: common goods account for 31 percent of aggregate food consumption in the richest decile, 45 percent in the median decile, and 61 percent in the poorest decile.

We next run the following regression to examine the cross-sectional variation in retailer market

²³In Section 6, we discuss the policy implications of this spatial heterogeneity for measuring real income inequality.

²⁴Note that DellaVigna and Gentzkow (2019) find lower prices in poorer areas, but the differences are economically small.

Table 3: Summary Statistics of the Main Sample (MSA-Quarter Level) by Income Deciles

	Decile 1	Decile 5	Decile 10
	Mean (SD)	Mean (SD)	Mean (SD)
Income per capita (\$ thousands)	31.615 (4.876)	39.010 (4.994)	56.733 (11.793)
Sales (\$ millions)	24.639 (18.431)	73.501 (74.623)	773.676 (748.232)
Population Share	0.001 (0.001)	0.002 (0.001)	0.019 (0.022)
Sales Share of Chains	0.152 (0.049)	0.132 (0.040)	0.084 (0.027)
Number of Chains	7.267 (2.440)	8.261 (2.447)	13.291 (4.901)
Number of Stores	58.788 (41.002)	91.243 (75.342)	535.664 (490.807)
Number of UPCs (thousands)	32.368 (12.131)	40.419 (13.292)	70.980 (22.167)

Note: The table provides the summary statistics of the main MSA-level sample for the aggregate food and beverages for three income-per-capita deciles: 1, 5, and 10. The population share is defined as the fraction of total population in an MSA. The sales share of chains is the average retailer-level sales share within an MSA, and the number of chains, stores, and UPCs is defined as the average number of chains, stores, and UPCs within an MSA.

structure across MSAs with different income levels:

$$Y_{mt} = \beta_0 + \beta_1 Income_{mt} + \delta_t + \varepsilon_{mt}, \quad (2)$$

where Y_{mt} is either sales, total count of chains or stores, the share of large retailers (defined by the top decile of total sales or store counts at the national level), or market concentration (measured by the HHI of retail chain sales) in MSA m in quarter t . $Income_{mt}$ is income per capita in MSA m in quarter t , and δ_t denotes quarter fixed effects.

The results in Table 4 reinforce the finding that poorer areas have lower sales, fewer retailers and stores, a higher fraction of large retailers, and higher market concentration. These results suggest that retailer market structure varies across regions with different income levels. In particular, poorer areas have more concentrated retail markets, with large firms accounting for a greater share of sales.

We show that the results are robust to alternative measures of market concentration: the sales share of the largest retailer (CR1) and the combined sales share of the three largest retailers (CR3)

Table 4: Retailer Market Structure across Regions with Different Income Levels

	Sales (in \$1mil.)	Chain#	Store#	Large Firm% (Sales)	Large Firm% (Store)	HHI
Income	26.37*** (5.876)	0.192*** (0.039)	16.43*** (3.928)	-0.002*** (0.001)	-0.009*** (0.002)	-0.004*** (0.001)
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	11,100	11,100	11,100	11,100	11,100	11,100

Note: The table presents regression results from equation (2), using the main sample in NielsenIQ. The coefficient of interest is on income per capita (in \$1000) in an MSA for a given quarter. The dependent variable is total sales in Column 1, the total count of chains in Column 2, the total count of stores in Column 3, the share of large retailers in Columns 4-5, where large firms are defined as the top decile of the national distribution of either total sales (Column 4) or the number of stores (Column 5), and the sales HHI in Column 6. Quarter fixed effects are included. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

within each MSA. The results are reported in Online Appendix D.1. We further examine the long-run cross-sectional relationship using MSA-level averages over the sample period. The long-run patterns are consistent with the baseline findings, as presented in Online Appendix D.2. We also repeat the analysis using the U.S. Census Bureau’s Business Dynamics Statistics (BDS), defining retailer size based on employment rather than sales. The resulting estimates are similar to our baseline findings, providing external validation of the relationship documented in the NielsenIQ data (Online Appendix D.3).

Finally, a potential concern is that measured market concentration in NielsenIQ may be affected by differential sample coverage across MSAs. For example, if NielsenIQ systematically provides more comprehensive retailer coverage in higher-income MSAs, differences in measured market concentration across regions could partly reflect sampling rather than underlying market structure. In Online Appendix E, we compare NielsenIQ retailer coverage with County Business Patterns (CBP) data from the U.S. Census Bureau and examine whether NielsenIQ coverage varies systematically with MSA income. We find no evidence that NielsenIQ coverage relative to CBP is systematically higher in higher-income MSAs.

3.3 Relationship between Inflation and Market Concentration

To explore spatial heterogeneity in inflation across regions with different degrees of market concentration, we estimate the following regression:

$$P_{mt} = \beta_0 + \beta_1 HHI_{mt} + \delta_t + \varepsilon_{mt}, \quad (3)$$

where P_{mt} is the geometric Laspeyres inflation rate of aggregate food in MSA m in quarter t .²⁵ HHI_{mt} is the HHI of retailer sales in MSA m in quarter t , and δ_t is a quarter fixed effect.

Table 5 reports estimates of equation (3). Column 1 shows that greater retailer market concentration is associated with higher food inflation. Column 2 adds income per capita as a control. Its coefficient is negative and statistically significant at the 10 percent level, consistent with poorer MSAs experiencing higher inflation. Column 3 additionally controls for annual MSA population, and Column 4 further adds the total number of retail chains. Across these specifications, the coefficient on HHI remains positive and statistically significant at the 1 percent level, ranging from 0.23 to 0.27. The stability of this association suggests that retailer market concentration may help account for spatial inflation differentials.²⁶

We conduct additional robustness checks using alternative measures of market concentration and specifications based on long-run averages. We also use the level of the geometric Laspeyres price index, rather than the quarterly inflation rate, to examine cumulative price differences across MSAs. The corresponding results are reported in Online Appendices F.1, F.2, and F.3, and are consistent with the baseline findings. Finally, we show that the same pattern holds in the egg market in Online Appendix F.4, which motivates the causal analysis in Section 4.

4 Causal Link between Inflation and Market Concentration

To study the causal relationship between inflation and retail market concentration, we focus on the egg market. In particular, we exploit a novel quasi-experiment based on the 2014-2015 avian

²⁵When constructing the aggregate food index at the MSA level, we use direct bilateral geometric Laspeyres indexes for two of the 21 component items—sugar and fruit—because their chained indexes exhibit substantial chain drift.

²⁶Income per capita enters negatively throughout, although its magnitude and statistical significance vary across specifications.

Table 5: Food Inflation across Regions

	Inflation	Inflation	Inflation	Inflation
HHI	0.251*** (0.039)	0.236*** (0.040)	0.225*** (0.039)	0.273*** (0.043)
Income		-0.002* (0.001)	-0.001 (0.001)	-0.002** (0.001)
Population			-0.001 (0.001)	-0.002*** (0.001)
Chain #				0.015*** (0.003)
Quarter FE	Yes	Yes	Yes	Yes
Observations	10,730	10,730	10,730	10,730

Note: The table presents regression results from equation (3), using the main sample in NielsenIQ. The coefficient of interest is on HHI in an MSA for a given quarter. The dependent variable is the geometric Laspeyres inflation rate (%) of aggregate food in an MSA for a given quarter. MSA-level income per capita (in \$1,000s) is included as a control starting in the second column, MSA population (in 100,000s), measured annually using BEA data, is included as an additional control in the last two columns. In addition, the total number of chains is included in the final column. Quarter fixed effects are included in all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

influenza outbreak and implement a triple-difference design to estimate heterogeneous pass-through across MSAs with different levels of concentration.

4.1 Pass-through of Bird Flu Shock in Egg Market

We use the 2014-2015 highly pathogenic avian influenza (bird flu) outbreak as an exogenous supply shock to the egg market. The 2014-2015 bird flu episode started in 2014Q4 and affected the prices and quantities of eggs sold in this period. Based on U.S. Department of Agriculture (USDA) reports, 36 million layers (birds that lay eggs) were lost due to the bird flu by June 2015.²⁷ This reduction in egg supply caused a sharp spike in egg prices, as shown in Figure 3.

Importantly, reports from the USDA and the Government Accountability Office (GAO) indicate that the impact of the bird flu shock varied geospatially, primarily affecting the central and western U.S. Farmers in these regions were more affected than farmers in other regions in terms of their layers' vulnerability to the disease. We have access to official data from the USDA on the timing,

²⁷The USDA also compensated producers that had to cull their layers. Payment was based on "fair market" values as determined by USDA appraisers.

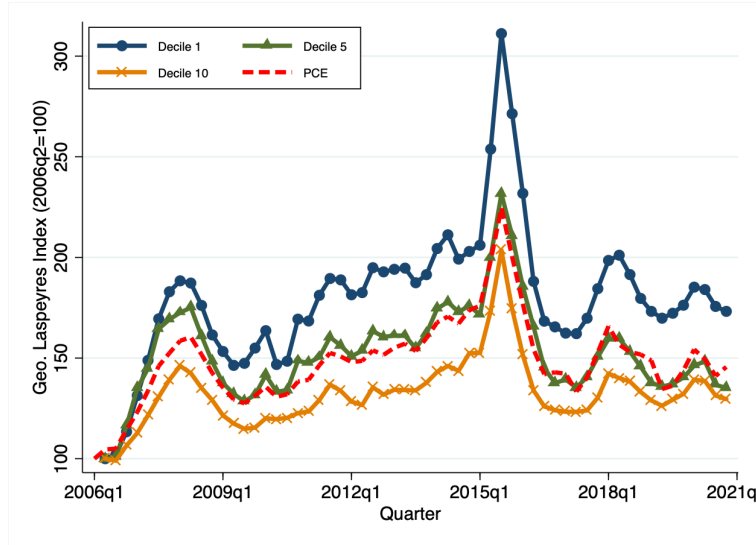


Figure 3: Price Index for Eggs

Notes: This figure represents the chained geometric Laspeyres price index constructed in NielsenIQ (solid lines) as well as the official personal consumption expenditures (PCE) index from the Bureau of Economic Analysis (dashed line) for eggs. The sample period runs from 2006Q2 to 2020Q3, and all series are normalized to the initial quarter. Each solid line corresponds to an income-per-capita decile of MSAs, where decile 1 includes MSAs with the lowest income per capita and decile 10 includes those with the highest income per capita. NielsenIQ UPCs are mapped to the PCE definition of food purchased for off-premises consumption using a product module concordance provided by the U.S. Bureau of Labor Statistics.

location, and number of bird layers that were culled.²⁸ By identifying MSAs where layers were culled, we can pinpoint areas disproportionately affected by the bird flu, which may have experienced higher inflation in egg prices early in the outbreak.^{29,30}

4.1.1 Difference-in-Differences Estimation

Using this information, we first implement a difference-in-differences design that compares egg-price inflation in treated and control MSAs before and after the bird flu outbreak.

To measure whether exposure of local farmers to culling affected local egg prices, we use a four-year window around the start of the bird flu episode in 2014Q4, and estimate the following

²⁸See details from the following Congressional report: <https://www.congress.gov/crs-product/R44114?hl=R44114&s=5&r=3>.

²⁹Based on the report, we identify the following list of impacted MSAs: Des Moines-Ames, Fargo-Valley City, Madison, Mankato, Minneapolis-St Paul, Omaha, Rochester-Madison City-Austin, Sioux City, Sioux Falls (Mitchell).

³⁰Egg markets are generally local or regional in nature. For example, Cal-Maine Foods, the largest producer and marketer of eggs in the U.S., primarily operates in the Southern region and had no facilities near the affected MSAs in 2015 (Online Appendix G).

Table 6: Difference-in-Differences Estimator

	Inflation	Inflation	Inflation	Abs. Inflation
Bird Flu $\times$ Post	-0.003 (0.004)	0.039*** (0.008)	-0.035*** (0.007)	0.053*** (0.006)
Sample Periods	All	Inflation	Deflation	All
Quarter FE	Yes	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes	Yes
Observations	3,145	1,850	1,295	3,145

Note: The table represents regression results from equation (4). The coefficient of interest is the interaction of Bird Flu and Post. Bird Flu is a binary variable that takes the value of one for MSAs in which egg farmers culled their layers during the 2014-2015 bird flu episode, and Post is a binary variable that takes the value of one in the post-shock period after 2014Q4. The sample period ranges from 2012Q4 to 2016Q4. Inflationary and deflationary periods are determined by the national price index of eggs. Inflation rate is the main outcome variable for Columns 1-3. Column 1 pools all periods together, Column 2 uses the inflationary period only, and Column 3 focuses on the deflationary period only. Column 4 pools all periods together and has the absolute value of the inflation rate as the outcome variable. MSA and quarter fixed effects are included. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

traditional two-way fixed effects regression from 2012Q4 to 2016Q4:

$$P_{mt} = \beta_0 + \beta_1(BirdFlu_m \times Post_t) + \delta_m + \delta_t + \varepsilon_{mt}, \quad (4)$$

where P_{mt} is the geometric Laspeyres inflation rate for eggs in MSA m in quarter t , $BirdFlu_m$ is an indicator variable equal to one for MSAs in which farmers culled their layers during the 2014-2015 bird flu outbreak, according to the USDA data, and $Post_t$ is a binary variable equal to one in the post-shock period (after 2014Q4). δ_m and δ_t are MSA and quarter fixed effects, respectively.

The results are shown in Table 6. In Column 1, we estimate a statistically insignificant (i.e., null) effect, which suggests that MSAs affected by the bird flu did not experience more aggregate egg price inflation. However, this null average effect masks substantial heterogeneity over time. When we separate the sample into inflationary and deflationary periods, when the national egg inflation rate is above or below zero, respectively, we observe opposing effects in the MSAs where layers were culled.³¹ In Column 2, we restrict the sample to the inflationary period when the national egg inflation rate was above zero. Here, we estimate a coefficient of 0.039 on the interaction of Bird Flu

³¹The inflationary period includes the following ten quarters: 2012Q4, 2013Q1, 2013Q2, 2013Q4, 2014Q1, 2014Q2, 2014Q4, and 2015Q1–Q3. The deflationary period includes the following seven quarters: 2013Q3, 2014Q3, 2015Q4, and 2016Q1–Q4.

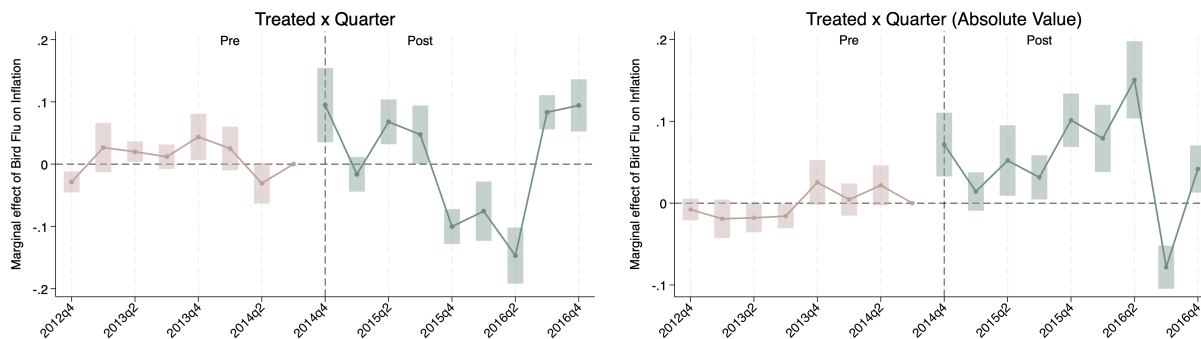


Figure 4: Event Study Difference-in-Differences (Bird Flu)

Notes: The figure presents the event-study difference-in-differences analysis of the differential effect of the 2014–2015 bird flu episode on egg inflation in MSAs whose farmers were directly affected. The outcome variable is egg price inflation in the left panel and its absolute value in the right panel. MSAs are assigned to the treatment group based on USDA reports identifying farms that culled layers. The post period begins in 2014Q4, with 2014Q3 as the reference quarter. Effects are estimated over the period 2012Q4–2016Q4. Standard errors are clustered at the MSA level.

and Post, which corresponds to a 3.9 percentage points higher egg inflation rate in MSAs affected by the bird flu after 2014Q4 during the inflationary period. This estimate is significant at the 1 percent level. In Column 3, we restrict the sample to the deflationary period and find that MSAs that culled their layers experienced a 3.5 percentage points lower inflation (higher deflation) rate after 2014Q4 during the deflationary period. This estimate is also significant at the 1% level. In Column 4, we pool all quarters and take the absolute value of the inflation rate as the main dependent variable. We find that MSAs that culled their layers experienced 5.3 percentage points larger absolute changes in the inflation rate after 2014Q4. This coefficient is significant at the 1 percent level.

These heterogeneous inflation effects during the bird flu outbreak are illustrated in Figure 4. The left panel presents the standard event-study difference-in-differences estimates, with the dashed vertical line marking 2014Q4 as the start of the post period. Prior to 2014Q4, there is no systematic difference in inflation between treated MSAs (those that culled layers) and control MSAs, indicating no evidence of differential pre-trends. After 2014Q4, treated MSAs exhibit higher inflation in the initial quarters and then lower inflation in subsequent quarters, yielding an average effect close to zero over the full post period.

These opposing inflation effects reflect heterogeneity across inflationary and deflationary regimes in the egg market. The right panel of Figure 4 illustrates this pattern by using the absolute value of inflation as the outcome variable. Treated MSAs are consistently more affected after 2014Q4,

while no significant differences are observed prior to the shock, further supporting the absence of differential pre-trends. The evidence indicates that the bird flu shock generated larger price increases in treated MSAs during the inflationary period and larger price declines during the deflationary period. This pattern holds in all post-period quarters except 2016Q3.

4.1.2 Triple-Difference Estimation

Next, we implement a triple-difference estimator to examine how the impact of the bird flu on egg inflation varies across treated MSAs with different degrees of market concentration (HHI). The following regression formalizes the identification strategy:

$$P_{mt} = \beta_0 + \beta_1 HHI_{mt} + \beta_2 (Treated_m \times Post_t) + \beta_3 (Treated_m \times HHI_{mt}) + \beta_4 (Post_t \times HHI_{mt}) + \beta_5 (Treated_m \times Post_t \times HHI_{mt}) + \delta_m + \delta_t + \varepsilon_{mt}, \quad (5)$$

where the subscript m corresponds to MSA m and subscript t corresponds to quarter t . $Treated_m$ is a binary variable indicating whether layers in MSA m were culled during the 2014-2015 bird flu episode according to the USDA report. $Post_t$ is a binary variable equal to 1 if quarter t is after 2014Q4 and zero otherwise. HHI_{mt} is the retailer-sales HHI for eggs in MSA m in quarter t . For the post-treatment period, we fix HHI to the pre-shock value in 2014Q3.³² P_{mt} is the geometric Laspeyres inflation rate for eggs in MSA m in quarter t . The fixed effect terms, δ_m and δ_t , are the same as before, and ε_{mt} is the error term.

The results are presented in Table 7. Column 1 pools all quarters within the four-year window and shows that treated MSAs with higher market concentration exhibit significantly higher inflation, on average, than those with lower concentration. The coefficient on the triple interaction implies that, for a one-unit increase in HHI, quarterly egg inflation rises by 5 percentage points among treated MSAs. This effect is statistically significant at the 1% level. Columns 2 and 3 decompose this effect into the inflationary and deflationary periods, respectively. In Column 2, restricting the sample to the inflationary period, we find that treated MSAs with higher market concentration experienced faster initial price increases in the egg market following the bird flu episode. This coefficient is also significant at the 1% level. In contrast, Column 3 shows that these MSAs did not

³²This is to avoid endogeneity concerns and to isolate the effect of supply shocks on local market concentration.

Table 7: Triple Difference Estimator (Eggs)

	Inflation	Inflation	Inflation
Bird Flu $\times$ Post $\times$ HHI	0.050*** (0.010)	0.084*** (0.019)	0.040* (0.023)
Bird Flu $\times$ Post	-0.030*** (0.007)	-0.006 (0.010)	-0.056*** (0.012)
HHI $\times$ Post	-0.010** (0.005)	-0.008 (0.008)	-0.008 (0.011)
Bird Flu $\times$ HHI	-0.165 (0.107)	-0.254 (0.156)	-0.075 (0.074)
HHI	0.037 (0.025)	0.056* (0.030)	0.026 (0.041)
Sample Periods	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
Observations	3,145	1,850	1,295

Note: The table represents regression results from equation (5). The coefficient of interest is the interaction of Bird Flu, Post, and HHI. Bird Flu is a binary variable that takes the value of one for MSAs in which egg farmers culled their layers during the 2014-2015 bird flu episode, and Post is a binary variable that takes the value of one in the post-shock period after 2014Q4. HHI is the Herfindahl-Hirschman index (HHI) of retail chains' sales of eggs within an MSA, fixed to 2014Q3 values for all quarters in the post period. The sample period ranges from 2012Q4 to 2016Q4. Inflationary and deflationary periods are determined by the national price index of eggs. Column 1 pools all periods together, Column 2 only considers the inflationary period, and Column 3 only considers the deflationary period. MSA and quarter fixed effects are included across all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

reduce prices by larger amounts during the deflationary period. We find that MSAs with higher concentration were slower to decrease prices, as indicated by a positive coefficient on the triple interaction term. This coefficient is statistically significant at the 10% level.

4.2 Robustness Checks

We conduct several robustness checks for the main results. The findings are robust to using terciles of HHI (Online Appendix H.1), alternative measures of market concentration such as CR1 and CR3 (Online Appendix H.2), and the inclusion of neighboring MSAs around treated markets (Online Appendix H.3). Additionally, in Online Appendix H.4, we conduct a placebo test by replacing market concentration with income per capita in the triple-difference specification. The resulting

triple-interaction coefficient shows no robust association with inflation, suggesting that our main results are unlikely to reflect differential responses between higher- and lower-income MSAs. Relatedly, the estimates remain stable in two additional specifications: one adds a triple interaction involving income per capita to the baseline specification, while the other controls for cross-MSA differences in pre-shock price levels (Online Appendix [H.5](#)).³³

Another potential concern is that the higher inflation observed in poorer regions with greater market concentration may be partially driven by differences in consumption baskets. As discussed in Section 2, poorer areas have access to fewer products than richer ones. To address this issue, we re-estimate the triple-difference specification at a more granular level, using UPC–MSA–quarter observations, as follows:

$$\begin{aligned} \Delta \ln price_{umt} = & \beta_0 + \beta_1 HHI_{mt} + \beta_2 (Treated_m \times Post_t) + \beta_3 (Treated_m \times HHI_{mt}) \\ & + \beta_4 (Post_t \times HHI_{mt}) + \beta_5 (Treated_m \times Post_t \times HHI_{mt}) + \delta_m + \delta_t + \delta_u + \varepsilon_{umt}, \end{aligned} \quad (6)$$

where $\Delta \ln price_{umt}$ denotes the log change in the price of UPC u in MSA m between quarters $t - 1$ and t . We include UPC fixed effects, δ_u , to ensure that identification comes from price changes within identical products. All other variables are defined as in the baseline specification.

Table 8 presents consistent results. In Column 1, we pool all quarters within the sample period and find that MSAs with higher market concentration saw a 1.6 percent increase in egg product prices relative to those with lower concentration. This estimate is significant at the 1 percent level. In Column 2, we restrict the sample to the inflationary period and find that MSAs affected by the bird flu with higher market concentration experienced a 3.4 percent increase in egg product prices relative to similarly affected MSAs with lower market concentration. This estimate is statistically significant at the 1 percent level. In Column 3, we limit the analysis to the deflationary period and find no statistically significant effect, suggesting persistence in the inflationary response rather than symmetric price reversals. The UPC-level results are consistent with the main MSA-level findings in Table 7. The UPC-level evidence further strengthens the argument that the inflationary effects are driven by retailer market concentration rather than differences in consumption baskets.

An alternative explanation is variation in retailer composition across MSAs, as different types of

³³[Sangani \(2026\)](#) documents lower price levels in lower-income areas; these differences can generate systematic heterogeneity in pass-through.

Table 8: Triple Difference Estimator (UPC-level)

	$\Delta \ln \text{Price}$	$\Delta \ln \text{Price}$	$\Delta \ln \text{Price}$
Bird Flu $\times$ Post $\times$ HHI	0.016*** (0.006)	0.034*** (0.010)	-0.010 (0.015)
Bird Flu $\times$ Post	-0.009** (0.004)	-0.005 (0.006)	-0.014 (0.010)
HHI $\times$ Post	-0.007** (0.003)	-0.009 (0.006)	-0.004 (0.007)
Bird Flu $\times$ HHI	0.001 (0.061)	-0.049 (0.085)	-0.076 (0.088)
HHI	0.024 (0.015)	0.054** (0.024)	-0.019 (0.027)
Sample Periods	All	Inflation	Deflation
Quarter FE	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes
UPC FE	Yes	Yes	Yes
Observations	145,989	84,418	61,470

Note: The table represents regression results from our triple difference-in-differences at the UPC level. HHI is the Herfindahl-Hirschman index (HHI) of retail chain's sales of eggs within an MSA, fixed to 2014Q3 values for all quarters in the post period. All else remains the same as before, except that MSA, UPC, and quarter fixed effects are included across all specifications. Standard errors are clustered at the MSA level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

retailers may exhibit different degrees of cost pass-through. In Online Appendix H.6, we show that our results are not driven by retailer composition differences. In addition, we provide consistent evidence using fluctuations in commodity prices in the coffee market, following Sangani (2024) (Online Appendix H.7). Lastly, we provide evidence that helps rule out other cost-side channels, such as differential wage growth in the retail sector (Online Appendix H.8).

4.3 Back-of-the-Envelope

To assess whether the magnitudes estimated in Table 7 are economically meaningful, we perform a back-of-the-envelope calculation to quantify how much of the inflation gap between the poorest and richest deciles can be explained by our market concentration mechanism. We examine differences in egg inflation between the top and bottom income deciles during the inflationary period.

To measure the contribution of market concentration, we compute the following term:

$$\pi_{contribution} = \frac{q \cdot \beta \cdot HHI_{diff}}{\pi_{d1} - \pi_{d10}}, \quad (7)$$

where $\pi_{contribution}$ denotes the share of the inflation gap attributable to differences in retailer market concentration. The term q denotes the number of quarters in the inflationary period (2014Q4–2015Q3), which equals four. π_{d1} and π_{d10} denote cumulative egg inflation for the poorest and the richest decile, respectively, over the same period. Inflation rates are measured in decimal units in the calculation.³⁴ The coefficient β corresponds to the estimate on the triple interaction Bird Flu $\times$ Post $\times$ HHI from Table 7, Column 2. Finally, HHI_{diff} denotes the difference in retail market concentration (HHI) between the poorest and richest deciles as of 2014Q3.

The cumulative inflation gap in eggs between the poorest and richest deciles from 2014Q4 to 2015Q3 was 14.5 percentage points (0.145 in decimal units), which corresponds to the denominator of equation (7). Based on our triple-difference estimates and observed differences in retailer market concentration across deciles, we estimate that 9.3 percentage points of this gap are accounted for by differences in retailer market concentration. Consequently, we obtain a value of 64 percent for $\pi_{contribution}$, suggesting that a substantial portion of the observed inflation gap during the bird flu episode can be explained by variation in local retailer concentration.

5 Theoretical Framework

5.1 Environment

In this section, we develop a theoretical framework to identify the mechanisms underlying our empirical findings. Following Atkeson and Burstein (2008), we model an economy comprising multiple food sectors within an MSA and assume a two-level nested CES demand system. Within each sector, a finite number of retailers compete in prices under Bertrand competition.

³⁴Because inflation is measured quarterly in decimal units, cumulative inflation over the four-quarter period is approximated by summing predicted quarterly effects.

At the top level, aggregate consumption across sectors s is given by

$$Y_t = \left(\int_s Y_{st}^{\frac{\sigma-1}{\sigma}} ds \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1, \quad (8)$$

where σ denotes the elasticity of substitution across sectors. Within each sector s , there is a discrete number of retailers $\mathcal{N}_s$ supplying differentiated varieties. Consumption at the sectoral level is

$$Y_{st} = \left(\sum_{i \in \mathcal{N}_s} N_{it}^{\frac{\eta}{\rho}} y_{it}^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}, \quad \rho > 1, \quad (9)$$

where ρ denotes the elasticity of substitution across retailers within a sector, with $\rho > \sigma$, and N_{it} denotes firm i 's stock of customer capital, which acts as a demand shifter.

Given this demand structure, firm $i \in \mathcal{N}_s$ faces the following demand function:

$$y_{it} = Y_t \left(\frac{P_{st}}{P_t} \right)^{-\sigma} \left(\frac{p_{it}}{P_{st}} \right)^{-\rho} N_{it}^{\eta}, \quad (10)$$

where P_t and P_{st} denote the aggregate price index and the sectoral price index.

Customer capital evolves over time as follows. At the beginning of period t , firm i inherits an undepreciated stock $(1 - \delta)N_{it}$. After production and sales take place, customer capital accumulates as a function of the firm's within-sector market share and depreciates at rate δ as follows:³⁵

$$N_{it+1} = (1 - \delta)N_{it} + (1 - \delta)s_{it}, \quad (11)$$

where s_{it} is firm i 's within-sector market share, defined as

$$s_{it} = \frac{p_{it}y_{it}}{P_{st}Y_{st}} = \frac{N_{it}^{\eta}p_{it}^{1-\rho}}{\sum_{j \in \mathcal{N}_s} N_{jt}^{\eta}p_{jt}^{1-\rho}}. \quad (12)$$

The corresponding sectoral and aggregate price indexes are

$$P_{st} = \left(\sum_{i \in \mathcal{N}_s} N_{it}^{\eta} p_{it}^{1-\rho} \right)^{\frac{1}{1-\rho}}, \quad P_t = \left(\int_s P_{st}^{1-\sigma} ds \right)^{\frac{1}{1-\sigma}}. \quad (13)$$

³⁵This formulation follows Foster et al. (2016a), who model customer capital as accumulating with firm revenue. Also, it aligns with Kleshchelski and Vincent (2009), where customer acquisition is proportional to past market share.

5.2 Firms

Each firm produces with marginal cost c_{it} , which is decreasing in productivity z_{it} .³⁶ Firm productivity is drawn from a Pareto distribution with scale and shape parameters, z_{min} and γ , respectively.

Firm i chooses its price to maximize the present discounted value of profits,

$$V(c_{it}, N_{it}) = \max_{p_{it}} (p_{it} - c_{it}) y_{it} + \beta \mathbb{E}_t V(c_{it+1}, N_{it+1}), \quad (14)$$

subject to the customer-capital accumulation process (11). The optimal pricing policy follows

$$y_{it} + (p_{it} - c_{it}) \frac{\partial y_{it}}{\partial p_{it}} + \beta(1 - \delta) \mathbb{E}_t \left[\frac{\partial V_{it+1}}{\partial N_{it+1}} \right] \frac{\partial s_{it}}{\partial p_{it}} = 0, \quad (15)$$

which reflects a dynamic trade-off between higher current markups and the effect of current pricing on future customer capital through market share. The envelope condition is

$$\frac{\partial V_{it}}{\partial N_{it}} = (p_{it} - c_{it}) \frac{\partial y_{it}}{\partial N_{it}} + \beta(1 - \delta) \mathbb{E}_t \left[\frac{\partial V_{it+1}}{\partial N_{it+1}} \right] \left(1 + \frac{\partial s_{it}}{\partial N_{it}} \right), \quad (16)$$

which captures the marginal value of customer capital.

Equation (15) highlights that pricing decisions are forward-looking: firms internalize how higher prices reduce current market share and, in turn, future customer capital. The strength of this intertemporal trade-off depends on the marginal value of customer capital, which varies across firms.³⁷

5.3 Steady-State Equilibrium

A steady-state equilibrium is a set of prices, quantities, customer capital, and value functions, $\left\{ \{p_i, y_i, s_i, N_i, V_i\}_{i \in \mathcal{N}_s}, P_s, P \right\}$, such that the demand system, the customer-capital accumulation, and the associated price indexes (10)–(13), together with the firm optimality conditions (15)–(16), are satisfied, with normalized output $Y = 1$.

³⁶For example, under constant-returns-to-scale production with labor only and wage w_t , marginal cost is $c_{it} = \frac{w_t}{z_{it}}$.

³⁷This dynamic channel disappears if $\delta = 1$ or if $\eta = 0$, in which customer capital fully depreciates or does not shift demand, respectively. In either case, the problem collapses to the static environment in Atkeson and Burstein (2008).

5.4 Markups and Cost Pass-through

In equilibrium, firms with larger market shares set higher markups, as shown in Proposition 1. The intuition is that firms with larger market shares face a lower marginal value of customer capital, which reduces the cost of losing customers and allows them to set higher prices and markups. This mechanism is consistent with Atkeson and Burstein (2008), while our framework additionally incorporates customer capital dynamics.

Proposition 1. *Firm markups are increasing in firm market share.*

Proof: See Appendix A.1.

We define the aggregate markup in a sector as the ratio of total revenue to total variable cost:

$$\mu_t = \frac{\sum_i p_{it} y_{it}}{\sum_i c_{it} y_{it}}. \quad (17)$$

The next proposition shows that aggregate markups depend on the distribution of firm market shares and increase with market concentration.³⁸

Proposition 2. *Aggregate markups are increasing in market concentration.*

Proof: See Appendix A.2.

We next characterize the relationship between firm market share and the pass-through of cost shocks.

Proposition 3. *Firm pass-through is incomplete and increases with firm market share once market share is sufficiently high.*

Proof: See Appendix A.3.

Suppose a firm faces an increase in marginal cost. The price response reflects both a direct effect of higher cost and an indirect effect operating through endogenous markups. The magnitude of the indirect effect is heterogeneous across firms and endogenously determined by the market share and customer capital of firms. When a firm is a monopolist ($s_{it} = 1$), pass-through is complete since a price increase does not affect market share and leaves customer capital and markups unchanged. Here, only the direct cost effect remains. More generally, when $0 < s_{it} < 1$, a higher price reduces

³⁸We provide supporting evidence of this prediction in Online Appendix I.1.

market share and customer capital by increasing the firm’s relative price. The resulting decline in the optimal markup partially offsets the increase in marginal cost, generating incomplete pass-through.

This markup adjustment becomes weaker for firms with sufficiently large market shares. Large firms face a smaller decline in market share following a price increase because their importance in the aggregate price index dampens changes in relative prices. Moreover, their larger customer base makes customer capital less sensitive to marginal changes in market share. As a result, these firms adjust markups less in response to cost shocks, leading to higher pass-through. Consequently, firms with sufficiently high market shares exhibit higher pass-through than smaller firms, suggesting that market concentration can influence aggregate pass-through.³⁹

5.5 Pricing Dynamics and Market Concentration

Next, we simulate the model to study how aggregate price responses and pass-through to a cost shock vary with market concentration. The parameterization is as follows. We set $\beta = 0.99$ to match a quarterly interest rate of 1.2%; $\rho = 3.85$, consistent with the median cross-firm elasticity estimates in [Hottman et al. \(2016\)](#); and $\delta = 0.15$ as in [Gourio and Rudanko \(2014\)](#). We normalize the Pareto scale parameter to $z_{min} = 1$ and set the shape parameter to $\gamma = 5.49$ following [Carvalho and Grassi \(2019\)](#). We jointly calibrate $\sigma = 1.61$ and $\eta = 0.31$ to match the top and bottom deciles of the retailer-level sales share distribution within an MSA in the data. The corresponding moments are 0.43 and 0.01 in the data, which the model matches by generating 0.44 and 0.01, respectively.

We assume firms are hit by a common aggregate shock x_t , such as the bird flu outbreak, that raises marginal costs additively to $x_t + c_{it}$.⁴⁰ We assume the following aggregate shock process:

$$x_1 > 0, \quad x_t = \phi x_{t-1} \text{ (for } t \geq 2), \quad 0 < \phi < 1,$$

where we set $x_1 = 0.1$ and $\phi = 0.875$. For simplicity, firm productivity is assumed to be time-invariant, z_i , implying that the idiosyncratic component of marginal cost, c_i , is also time-invariant.

³⁹We provide supporting evidence of this prediction in Online Appendix [I.2](#).

⁴⁰Note that the common additive cost shock induces reallocation across firms within each economy as its proportional magnitude varies with firms’ initial marginal costs. In our cross-economy comparison, we hold each focal firm’s productivity—and hence its initial marginal cost and proportional cost change—fixed across economies with different levels of concentration. The comparison therefore isolates how market structure shapes pass-through, holding the firm’s cost level and cost change fixed.

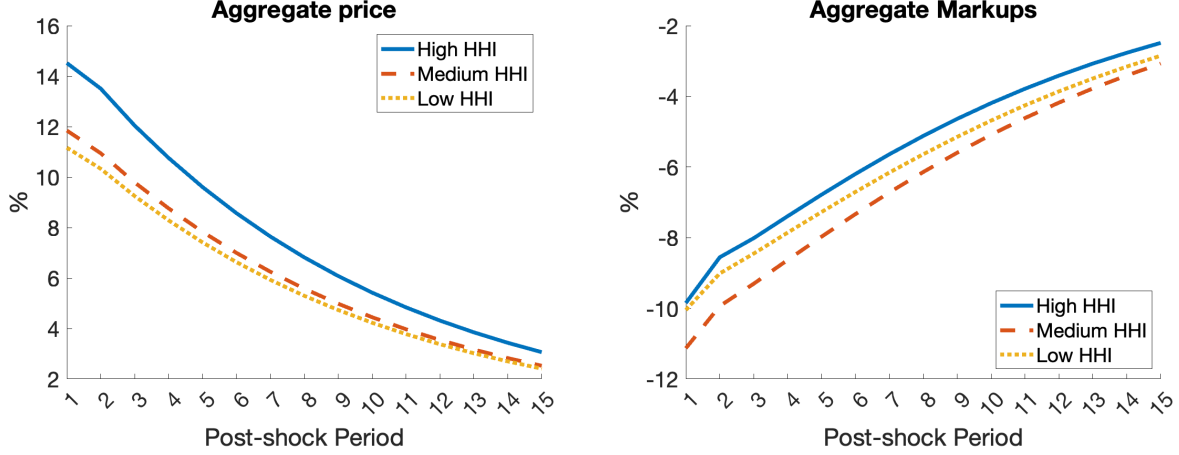


Figure 5: Response of Aggregate Price and Markups to a Cost Shock

Notes: The figure plots the impulse responses of the aggregate price level (left panel) and markups (right panel) to a positive common cost shock. “High HHI” corresponds to the economy with $\mathcal{N}_s = 3$, and “Medium HHI” and “Low HHI” denote the economies with $\mathcal{N}_s = 10$ and $\mathcal{N}_s = 17$, respectively. Aggregate prices are measured as the average of firm-level prices within each economy, and aggregate markups are constructed using equation (17).

In the baseline economy, the number of retailers is set to $\mathcal{N}_s = 10$, which corresponds to the average number of chains per MSA in our data. To discipline differences in market concentration, we vary $\mathcal{N}_s$ while holding all other structural parameters fixed.⁴¹ We compare the baseline economy to counterfactuals with $\mathcal{N}_s = 3$ and $\mathcal{N}_s = 17$, which correspond to two standard deviations below and above the mean, respectively. We find that market concentration declines with the number of retailers, even when the underlying productivity distribution is held fixed.

For each economy, we present the impulse response functions of aggregate prices and markups following a cost increase in Figure 5.⁴² Prices rise after the shock, and this increase is more pronounced in the high-HHI ($\mathcal{N}_s = 3$) economy. This difference persists along the transition path, as price changes remain higher in the high-HHI economy. These patterns are consistent with our empirical evidence on pass-through following cost shocks. Furthermore, aggregate markups decline on impact, but the magnitude of the decline is smaller in the high-HHI economy.

To examine these results more closely, we investigate the impulse response functions of firm-level prices for the largest and smallest firms in the baseline economy and compare them with their

⁴¹In each economy, firms are drawn from the same productivity distribution with identical parameters. This ensures that firms share the same fundamentals across economies, while their pricing decisions and outcomes may still vary with different market structure (the number of competitors).

⁴²This figure is based on a single simulation. The results are robust to Monte Carlo averaging across repeated simulations and to alternative numbers of simulation draws.

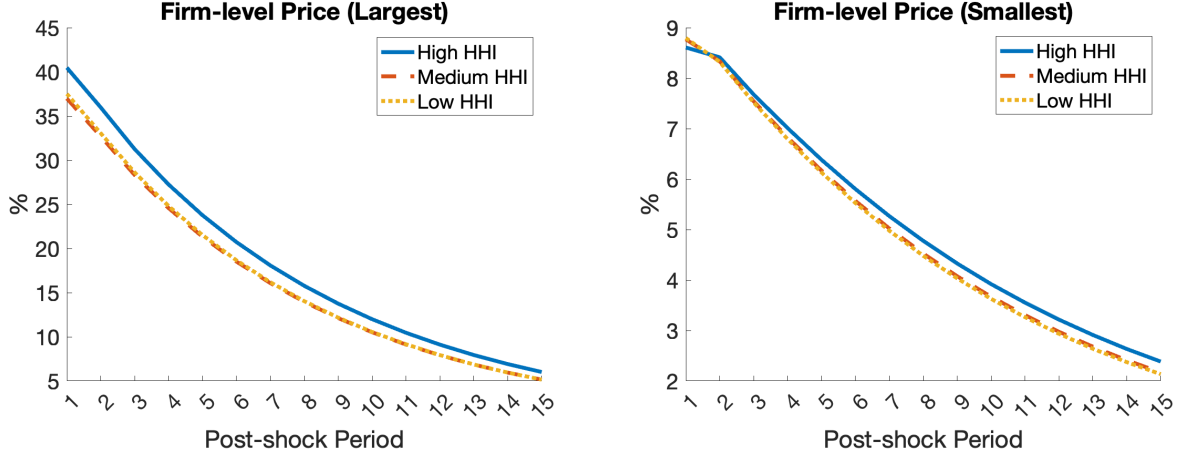


Figure 6: Response of Firm-Level Prices (Large vs. Small) to a Cost Shock

Notes: The figure plots the impulse responses of firm-level prices for the largest retailer (left panel) and the smallest retailer (right panel) in the baseline economy to a positive common cost shock. In the counterfactual economies, we use the corresponding firms with the same fundamentals as these two baseline firms. “High HHI” corresponds to the economy with $\mathcal{N}_s = 3$, and “Medium HHI” and “Low HHI” denote the economies with $\mathcal{N}_s = 10$ and $\mathcal{N}_s = 17$, respectively.

counterparts with the same productivity (and hence the same initial marginal cost and cost shock) in the other two economies.

Figure 6 presents the results, showing that the largest firm raises its price on impact, with the strongest response occurring in the high-HHI economy ($\mathcal{N}_s = 3$). This firm has the highest market share and the largest stock of customer capital relative to otherwise identical firms in lower-HHI economies ($\mathcal{N}_s = 10$ or 17). When the shock hits, the largest firm in the high-HHI economy can raise prices more aggressively as it faces lower sensitivity of sales share to price changes and a lower marginal value of customer capital. Consequently, it exhibits higher pass-through, as characterized in Proposition 3. Consistent with this mechanism, the declines in its sales share, markups, and customer capital are more muted than those in less concentrated economies, as shown in Figure 7.

By contrast, the responses of the smallest firms vary less across economies. Their sales shares are relatively similar, and their price and markup responses exhibit little variation across concentration levels. This suggests that aggregate dynamics are driven disproportionately by the behavior of large firms, whose responses vary more systematically with market concentration through the interaction of market share, demand sensitivity, and the value of customer capital.⁴³

⁴³Customer capital shapes this cross-sectional heterogeneity in pass-through through the feedback between current pricing and future customer accumulation. This channel is particularly important for large firms, whose equilibrium market share compounds through this channel and becomes more sensitive to market structure than under static pricing (Atkeson and Burstein, 2008). This pattern is largely absent under static pricing and is consistent with our empirical

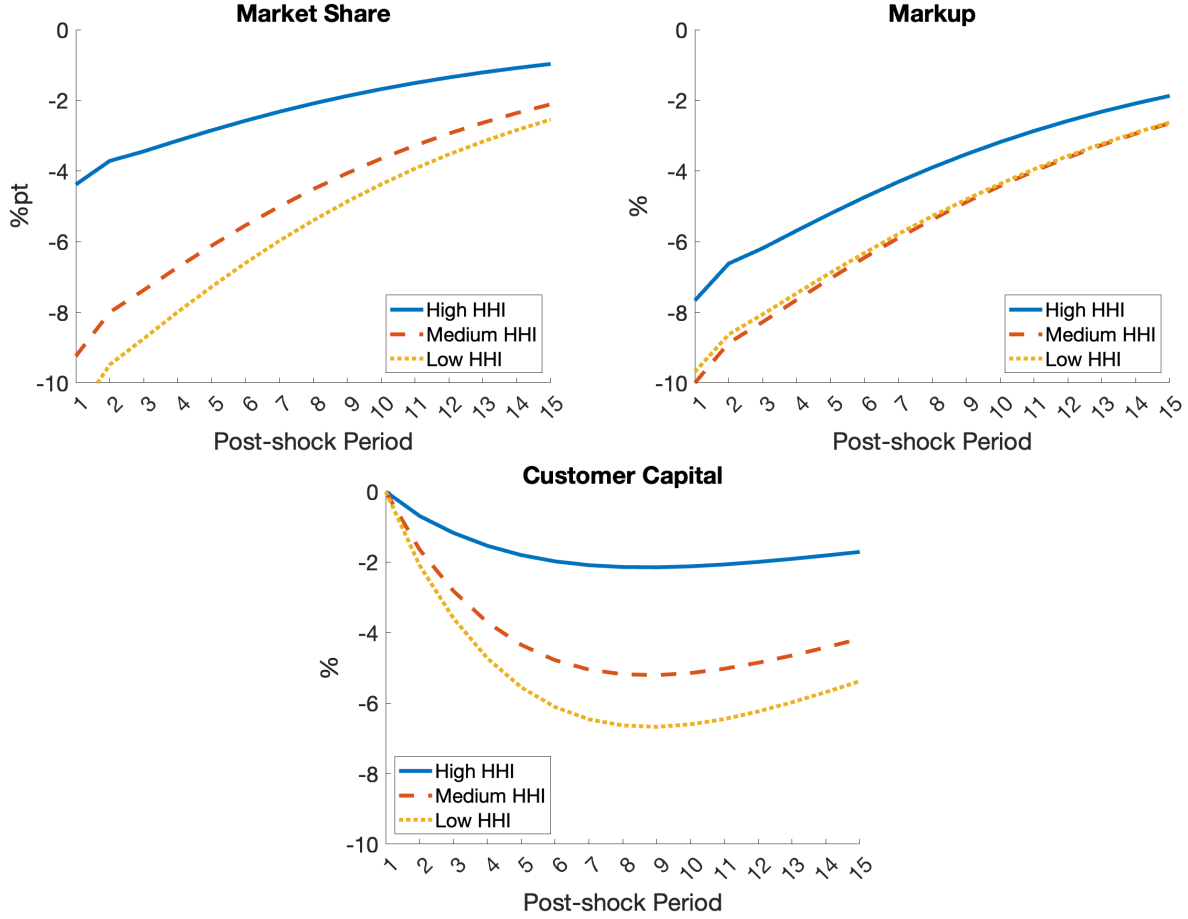


Figure 7: Response of the Largest Firm's Sales Share, Markup and Customer Capital to a Cost Shock

Notes: The figure plots the impulse responses of firm-level market share (top left), markups (top right), and customer capital (bottom panel) for the largest retailer in the baseline economy following a positive common cost shock. In the counterfactual economies, we consider the corresponding firm with the same fundamentals as the focal firm. “High HHI” corresponds to the economy with $\mathcal{N}_s = 3$, and “Medium HHI” and “Low HHI” denote the economies with $\mathcal{N}_s = 10$ and $\mathcal{N}_s = 17$, respectively.

The model highlights market structure as a potential amplifier of the welfare costs of cost shocks. High-HHI economies exhibit higher pass-through, which is accompanied by higher equilibrium markups (Proposition 2). As a result, market concentration can exacerbate welfare losses associated with market power in the presence of cost shocks. In the current framework, the number of firms is taken as given and determines the degree of market concentration. Endogenizing firm entry and market structure would allow an explicit analysis of optimal policy and may reveal welfare gains from policies that promote entry and competition. This is an avenue we leave for future work.

evidence in Online Appendix I.2. This also gives a more pronounced aggregate price response to cost shocks in concentrated markets than the static setup does.

6 Policy Implications

Our findings have implications for how policymakers evaluate the incidence of food supply shocks and measure real purchasing power across regions. First, our results show that food affordability depends not only on the size of food supply shocks, but also on local market structure. Food affordability has become one of the most salient economic issues in the U.S.⁴⁴ Although we study the 2014-2015 avian influenza episode, food products have continued to experience recurring cost shocks. Subsequent bird flu outbreaks have again driven up egg prices; the spread of the New World screwworm has raised beef prices by reducing cattle herds; and broader supply disruptions, including the 2026 oil price shock, have placed additional upward pressure on food prices. Our findings therefore have implications that extend beyond this particular episode.

In particular, our results imply that the same aggregate food supply shock can have different inflationary effects across local markets because of differences in local market structure. While localized supply disruptions can generate larger price increases than national statistics suggest (Butters et al., 2022), we show that even identical cost shocks are transmitted differently across markets. Pass-through is systematically higher in more concentrated grocery markets, so repeated food supply shocks can generate persistent spatial differences in food inflation. Because grocery market concentration is greater in many lower-income areas, these shocks may disproportionately erode food affordability in places where households are least able to absorb higher prices.

Second, our findings have implications for measuring real income inequality. Official price indexes are typically available only at the national level or for relatively wealthy MSAs.⁴⁵ Our results show that food cost shocks are transmitted unevenly across local markets because pass-through depends on local grocery market structure. As a result, households in more concentrated markets—which are typically lower-income areas—experience systematically higher food inflation following supply shocks. Yet national price indexes are expenditure-weighted averages across regions, which primarily reflect higher-income areas.⁴⁶ Relying on national price indexes can therefore overstate purchasing power in lower-income regions and bias measures of real income inequality and regional cost-of-living differences.⁴⁷ Appendix B documents this implication by

⁴⁴Food prices featured prominently in both the 2024 presidential debates and the 2026 State of the Union address.

⁴⁵See details at <https://www.bls.gov/cpi/regional-resources.htm>.

⁴⁶Figure 1 illustrates this pattern.

⁴⁷For example, if nominal wage growth were identical across regions, national price indexes would overstate real

comparing regional real income gaps measured using national and MSA-specific food price indexes.

Lastly, our results suggest that market power can amplify consumer welfare losses following cost shocks, highlighting a role for competition policy beyond its effects on static markups. The substantial spatial heterogeneity in market concentration further implies that the incidence of cost shocks may differ systematically across regions, particularly between low- and high-income regions. This provides a rationale for incorporating local market conditions more explicitly into competition policy, through geographically targeted merger scrutiny, entry-promoting policies, or other place-based interventions. More broadly, place-based transfers or market regulation may help mitigate the unequal burden of cost shocks when disparities in local competitive conditions cannot be readily addressed through conventional antitrust policy.⁴⁸

7 Concluding Remarks

In this paper, we investigate spatial variation in food inflation and the role of retailer market structure. Using NielsenIQ data, we document that lower-income MSAs experience systematically higher food inflation than wealthier areas. Poorer MSAs also exhibit fewer product varieties and retailers, a higher share of large chains, and greater market concentration. Exploiting an exogenous cost shock from the 2014-2015 bird flu outbreak, we show that MSAs with higher retail concentration experience larger inflation spikes and slower subsequent price declines. This implies that temporary shocks generate persistent differences in local inflation, thereby contributing to narrower price-level gaps while widening real-income gaps between poorer and richer regions. We provide a model in which large retailers' market power and customer capital amplify cost pass-through in more concentrated markets. Our findings highlight the central role of local market structure in shaping inflation dynamics, consumer welfare, and real income inequality. They also underscore the limitations of national inflation measures, which can mask economically meaningful local price variation, and provide policy-relevant insights for the design of antitrust and place-based policies.

wage growth in poorer areas because they fail to account for their systematically higher food inflation.

⁴⁸National concentration measures can obscure local effects: a merger with only modest effects on national concentration may substantially increase concentration in particular local markets.

Appendix

A Analytical Results and Proofs

A.1 Proof of Proposition 1

Proof. The price elasticity of demand for firm i , ϵ_{it} , can be derived as follows. Using the demand curve (10),

$$\ln y_{it} = \ln Y_t - \sigma \ln \frac{P_{st}}{P_t} - \rho \ln \frac{p_{it}}{P_{st}} + \eta \ln N_{it}. \quad (\text{A.1})$$

Then, the elasticity ϵ_{it} is

$$\epsilon_{it} \equiv -\frac{\partial \ln y_{it}}{\partial \ln p_{it}} = -(\rho - \sigma) \frac{\partial \ln P_{st}}{\partial \ln p_{it}} + \rho. \quad (\text{A.2})$$

With (13), it follows that $\frac{\partial P_{st}}{\partial p_{it}} = N_{it}^\eta \left(\frac{p_{it}}{P_{st}}\right)^{-\rho}$, and therefore

$$\frac{\partial \ln P_{st}}{\partial \ln p_{it}} = \frac{p_{it}}{P_{st}} \frac{\partial P_{st}}{\partial p_{it}} = N_{it}^\eta \left(\frac{p_{it}}{P_{st}}\right)^{1-\rho} = s_{it}. \quad (\text{A.3})$$

Thus, the effect of a firm's price on the sectoral price index depends on its market share. The larger the firm's share, the greater its influence on the sectoral price index.

Rephrasing (A.2) with (A.3), we obtain the following relationship,

$$\epsilon_{it} \equiv (1 - s_{it})\rho + s_{it}\sigma, \quad (\text{A.4})$$

showing that the elasticity is a linear function of firm market share. Since $\rho > \sigma$, the demand elasticity declines with the firm's market share.

Combining the pricing condition (15) and the envelope condition (16) at the steady state yields

$$\frac{p_{it} - c_{it}}{p_{it}} = \frac{1}{\hat{\epsilon}_{it}}, \quad (\text{A.5})$$

where

$$\hat{\epsilon}_{it} \equiv \epsilon_{it} + \frac{\beta \delta \eta (\rho - 1) (1 - s_{it}) \left(1 - \frac{\rho - \sigma}{\rho - 1} s_{it}\right)}{1 - \beta [(1 - \delta) + \eta \delta (1 - s_{it})]}. \quad (\text{A.6})$$

Hence the firm's markup is

$$\mu_{it} = \frac{p_{it}}{c_{it}} = \frac{\hat{\epsilon}_{it}}{\hat{\epsilon}_{it} - 1}. \quad (\text{A.7})$$

Note that $\hat{\epsilon}_{it}$ denotes the effective elasticity faced by the firm. The second term in $\hat{\epsilon}_{it}$ captures the additional dynamic effect arising from customer capital accumulation, which is decreasing in the firm's market share.⁴⁹ For expositional convenience, let $A \equiv 1 - \beta(1 - \delta)$ and $B \equiv \beta\eta\delta$, then we obtain

$$\frac{d\hat{\epsilon}_{it}}{ds_{it}} = \frac{-A((\rho - \sigma)A + B(\sigma - 1))}{(A - B(1 - s_{it}))^2} < 0.$$

This, along with equation (A.6), proves that firm markups are increasing in firm market share as follows:

$$\frac{d\mu_{it}}{ds_{it}} = \frac{d\mu_{it}}{d\hat{\epsilon}_{it}} \frac{d\hat{\epsilon}_{it}}{ds_{it}} = -\frac{1}{(\hat{\epsilon}_{it} - 1)^2} \frac{d\hat{\epsilon}_{it}}{ds_{it}} > 0. \quad (\text{A.8})$$

□

A.2 Proof of Proposition 2

Proof. First, at the stationary equilibrium, markup follows (A.7). We can rewrite it as:

$$\mu_{it} = \frac{1}{1 - \frac{1}{\hat{\epsilon}(s_{it})}}, \text{ where } \hat{\epsilon}(s_{it}) \equiv \rho - (\rho - \sigma)s_{it} + \frac{\beta\delta\eta(\rho - 1)(1 - s_{it}) \left(1 - \frac{\rho - \sigma}{\rho - 1}s_{it}\right)}{1 - \beta[(1 - \delta) + \eta\delta(1 - s_{it})]}. \quad (\text{A.9})$$

Recall that aggregate markup is

$$\mu_t = \left(\sum_i \frac{s_{it}}{\mu_{it}} \right)^{-1} = \left(\sum_i s_{it} \left(1 - \frac{1}{\hat{\epsilon}(s_{it})} \right) \right)^{-1}. \quad (\text{A.10})$$

Following [Alviarez et al. \(2025\)](#) and [Grassi et al. \(2017\)](#), we approximate the relationship around $s_{it} = 0$. Then

$$\frac{1}{\hat{\epsilon}(s_{it})} \simeq \frac{1}{\hat{\epsilon}(0)} + \kappa s_{it},$$

⁴⁹Intuitively, customer capital increases effective demand elasticity more (less) strongly for low(high)-market-share firms. Thus, low-market-share firms have a stronger incentive to restrain current prices in order to preserve and accumulate future customers, whereas this incentive is weaker for high-market-share firms.

where $\kappa \equiv -\frac{\hat{\epsilon}'(0)}{\hat{\epsilon}(0)^2}$, so that

$$\frac{s_{it}}{\mu_{it}} \simeq s_{it} \left(1 - \frac{1}{\hat{\epsilon}(0)} - \kappa s_{it} \right). \quad (\text{A.11})$$

Summing across firms and using $\sum_i s_{it} = 1$ gives

$$\sum_i \frac{s_{it}}{\mu_{it}} \simeq 1 - \frac{1}{\hat{\epsilon}(0)} - \kappa \sum_i s_{it}^2 = 1 - \frac{1}{\hat{\epsilon}(0)} - \kappa HHI_t. \quad (\text{A.12})$$

Hence aggregate markup is approximately

$$\mu_t \simeq \frac{1}{1 - \frac{1}{\hat{\epsilon}(0)} - \kappa HHI_t}. \quad (\text{A.13})$$

As $\kappa \equiv -\frac{\hat{\epsilon}'(0)}{\hat{\epsilon}(0)^2} > 0$, this confirms that aggregate markup increases with the HHI. $\square$

A.3 Proof of Proposition 3

Proof. Let $\lambda_{it} \equiv \frac{d \ln p_{it}}{d \ln c_{it}}$. Differentiating $p_{it} = \mu_{it} c_{it}$ gives

$$\lambda_{it} = \frac{d \ln p_{it}}{d \ln c_{it}} = 1 + \frac{d \ln \mu_{it}}{d \ln c_{it}}. \quad (\text{A.14})$$

Note that with no markup, there will be complete pass-through

$$\frac{d \ln p_{it}}{d \ln c_{it}} = 1.$$

If not, in more general cases in which $\frac{d \ln \mu_{it}}{d \ln c_{it}} \neq 0$, we obtain

$$\frac{d \ln \mu_{it}}{d \ln c_{it}} = \frac{d \ln \mu_{it}}{ds_{it}} \cdot \frac{ds_{it}}{d \ln c_{it}} = \underbrace{\frac{d \ln \mu_{it}}{d \hat{\epsilon}_{it}}}_{-\frac{1}{\hat{\epsilon}_{it}(\hat{\epsilon}_{it}-1)} < 0} \cdot \underbrace{\frac{d \hat{\epsilon}_{it}}{ds_{it}}}_{< 0} \cdot \frac{ds_{it}}{d \ln c_{it}} = \underbrace{\frac{-\hat{\epsilon}'(s_{it})}{\hat{\epsilon}_{it}(\hat{\epsilon}_{it}-1)}}_{> 0} \cdot \frac{ds_{it}}{d \ln c_{it}}. \quad (\text{A.15})$$

This implies that the sign depends on $\frac{ds_{it}}{d \ln c_{it}}$. Further expanding this term

$$\frac{ds_{it}}{d \ln c_{it}} = \frac{ds_{it}}{d \ln p_{it}} \frac{d \ln p_{it}}{d \ln c_{it}}.$$

With (12), we know that

$$\ln s_{it} = \eta \ln N_{it} + (1 - \rho) \ln p_{it} - (1 - \rho) \ln P_{st}$$

and

$$\frac{d \ln s_{it}}{d \ln p_{it}} = (1 - \rho) - (1 - \rho) s_{it} = (1 - \rho)(1 - s_{it}),$$

as $\frac{d \ln P_{st}}{d \ln p_{it}} = s_{it}$. Substituting this expression above,

$$\frac{ds_{it}}{d \ln c_{it}} = \frac{ds_{it}}{d \ln p_{it}} \frac{d \ln p_{it}}{d \ln c_{it}} = s_{it}(1 - \rho)(1 - s_{it}) \frac{d \ln p_{it}}{d \ln c_{it}}.$$

Then combining it with (A.15) and plugging them into (A.14), we have the pass-through of a cost-shock to prices as follows:

$$\lambda_{it} = \frac{d \ln p_{it}}{d \ln c_{it}} = \frac{1}{1 + (\rho - 1) s_{it}(1 - s_{it}) \frac{-\hat{\epsilon}'(s_{it})}{\hat{\epsilon}_{it}(\hat{\epsilon}_{it} - 1)}}, \quad (\text{A.16})$$

where $\hat{\epsilon} = \hat{\epsilon}(s_{it})$ following (A.6). Given that $\rho > 1$, $0 < s_{it} < 1$, and $\hat{\epsilon}'(s_{it}) < 0$,

$$0 < \lambda_{it} = \frac{d \ln p_{it}}{d \ln c_{it}} < 1,$$

establishing positive but incomplete pass-through.

Next, we prove how pass-through depends on firms' market shares. Let $\Phi_{it} \equiv (\rho - 1) s_{it}(1 - s_{it}) \frac{-\hat{\epsilon}'(s_{it})}{\hat{\epsilon}_{it}(\hat{\epsilon}_{it} - 1)}$ so that

$$\lambda_{it} = \frac{1}{1 + \Phi_{it}},$$

then we want to verify the sign of $\frac{\partial \Phi_{it}}{\partial s_{it}}$, and thus, the sign of $\frac{\partial \lambda_{it}}{\partial s_{it}} = -\frac{1}{(1 + \Phi_{it})^2} \frac{\partial \Phi_{it}}{\partial s_{it}}$.

For expositional convenience, let $A \equiv 1 - \beta(1 - \delta)$, $B \equiv \beta\eta\delta$, $\Gamma \equiv A((\rho - \sigma)A + B(\sigma - 1))$, $U(s_{it}) \equiv (\rho A - B - ((\rho - \sigma)A - B) s_{it})$, and $V(s_{it}) \equiv (\rho A - B - ((\rho - \sigma)A - B) s_{it} - A + B(1 - s_{it}))$. Then, we can simplify Φ_{it} as follows,

$$\Phi_{it} = (\rho - 1) s_{it}(1 - s_{it}) \frac{\Gamma}{U(s_{it})V(s_{it})},$$

which gives

$$\begin{aligned}\frac{\partial \Phi(s_{it})}{\partial s_{it}} &= \frac{(\rho - 1)\Gamma}{U(s_{it})^2 V(s_{it})^2} \left((\rho A - B)(\rho - 1)A(1 - s_{it})^2 - \sigma(\sigma - 1)A^2 s_{it}^2 \right) \\ &= \frac{(\rho - 1)\Gamma}{U(s_{it})^2 V(s_{it})^2} \left(\sqrt{(\rho A - B)(\rho - 1)A}(1 - s_{it}) + \sqrt{\sigma(\sigma - 1)A^2 s_{it}} \right) \\ &\quad \cdot \left(\sqrt{(\rho A - B)(\rho - 1)A}(1 - s_{it}) - \sqrt{\sigma(\sigma - 1)A^2 s_{it}} \right).\end{aligned}$$

Note that the first two terms are strictly positive, so the sign of $\frac{\partial \Phi(s_{it})}{\partial s_{it}}$ (and thus of $\frac{\partial \lambda_{it}}{\partial s_{it}}$) is determined by the last term. This implies that, for a sufficiently high market share above the threshold $s^* = \frac{\sqrt{(\rho A - B)(\rho - 1)A}}{\sqrt{(\rho A - B)(\rho - 1)A} + \sqrt{\sigma(\sigma - 1)A^2}}$,

$$\frac{\partial \Phi(s_{it})}{\partial s_{it}} < 0 \quad \text{for sufficiently large } s_{it} > s^*.$$

Consequently,

$$\frac{\partial \lambda_{it}}{\partial s_{it}} > 0 \quad \text{for sufficiently large } s_{it} > s^*.$$

Therefore, pass-through is positive and incomplete for firms with $0 < s_{it} < 1$, and increases with firm market share once market share is sufficiently high. $\square$

B Real Income Inequality

We examine nominal and real income per capita across income deciles over time. We focus on food-at-home given the coverage of the NielsenIQ data. First, we estimate the share of nominal income per capita allocated to food for each MSA. The Consumer Expenditure Survey (CEX) reports food-at-home expenditure shares by household income ranges, with nine groups defined prior to 2014 and seven groups thereafter. Using the NielsenIQ Consumer Panel, we map household income into these groups and compute each group's expenditure weight within an MSA. We then construct the MSA-level food expenditure share as the weighted average of CEX group-level food shares using these weights. Multiplying by MSA-level nominal income per capita from the Bureau of Economic Analysis yields average income allocated to food at the MSA level. Next, we average this measure across MSAs within each decile for the left panel. To obtain real income per capita, we

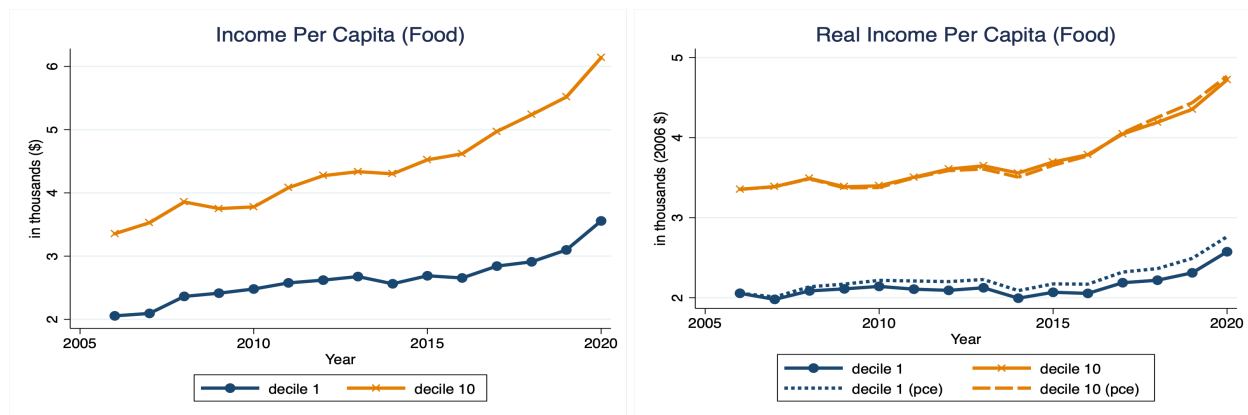


Figure B.1: Nominal and Real Income per Capita

Notes: This figure displays nominal and real income per capita allocated to food (in thousands of dollars) across income deciles, with the left panel showing nominal values and the right panel showing real values (food price deflator). For the two solid lines in the right panel, real income is constructed using decile-level food price indexes derived from NielsenIQ. The two dashed lines use the official PCE food price index for deflation, where the short-dashed line corresponds to decile 1 and the long-dashed line to decile 10. The sample period is 2006 to 2020. Decile 1 represents MSAs in the bottom income decile, and decile 10 represents those in the top income decile.

deflate the nominal series using the food price index constructed from the NielsenIQ Retail Scanner.

The two solid lines in Figure B.1 illustrate the patterns of nominal and real income series for the top (decile 10) and bottom (decile 1) income deciles. The data indicate that both nominal and real income per capita have generally increased across all deciles. In particular, the gap between the top and bottom deciles has widened over time for both measures.

Ignoring differences in regional inflation compresses measured real income gaps. This is illustrated by the two dashed lines in the right panel of Figure B.1, which plot real income per capita for the bottom and top deciles constructed using the official PCE food price index. The figure shows that relying on this national index overstates real income in decile 1, both in levels and in growth rates, but has little effect on the decile 10 series relative to the baseline. As a result, it understates the real income gap and, in turn, real income inequality.

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